

Convex Optimization

Ruhi Sonal

Department of Social Sciences and Humanities

IIITD

Google classroom code: **pzk7th4**

Indicative Solutions

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Review and practice

- Consider the following optimization problem: [Q1, PS4]

$$\max\{x^2 - y^2 : x^2 + y^2 = 1\}$$

- Is the above problem convex? No.
- Steps to maximize: Check KKT necessary (first order conditions)
- Find points using KKT sufficiency.

$$\textcircled{1} \quad \nabla f_0(x^*) + \sum_{i \in M} \lambda_i \nabla f_i(x^*) = 0$$

$$\textcircled{2} \quad \lambda_i f_i(x^*) = 0, \forall i \in M$$

$$\textcircled{3} \quad \lambda_i \geq 0 \quad \forall i \in M$$

$$\textcircled{4} \quad f_i(x^*) \geq 0, \forall i \in M.$$

Review: convexity and conic structures

- Difference between convexity of a set vs convexity of functions.
- Is a conic hull always convex? Yes.
- Is a convex hull always a cone? No.
- Is a cone an affine set? No.

Existence of maxima/minima

- Let $f : R^n \rightarrow R$ be a convex, continuously differentiable function. Consider a compact subset $D \subset R$. Does f have a maxima/minima in D ? If yes, is it a fixed point? Yes. **Not necessarily a fixed point- if x^* lies on the boundary of D , it may not be a fixed point. Recollect first order condition can be rewritten as fixed point at x^* .**

Definition 1

(Weierstrass maximum Theorem) Let $S \subset R^n$ be compact and f a continuous real valued function on S . Then $\arg \max_{x \in S} f(x)$ and $\arg \min_{x \in S} f(x)$ both exist in S .

KKT conditions: Q2 [PS 5]

- KKT conditions are sufficient for the existence of an optimal point for a convex optimization problem. True/False? provide a formal explanation in support of your answer.
- Ans. Sufficient because the problem can be written as concave dual function. See lecture 11-13 for details. The conditions are necessary when ∇f_i are linearly independent.

Proportional division and envy free-ness [Q3, PS 5]

- Agents do not have any incentive to stop at $1/n$. If they did, it would be envy free and proportional. In general:
- No. It is neither proportional nor envy free. Agent 1 has incentive to not say stop until $y = 1$. In this case $v_2(x_2), v_3(x_3) = 0 < \frac{1}{n}$. 2, 3 will prefer x_1 over their allocations.
- Can we make some changes to ensure proportionality? If instead of agent 1, any agent can say “stop” at any y and gets the piece $[0, y]$.

Integer programs [Q4, PS 5]

- Is the following matrix TUM?
$$\begin{bmatrix} 1 & -1 & -1 & 0 \\ -1 & 0 & 0 & 1 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

If yes, write down a set of constraints that are represented by the above matrix for an integer programming problem. If no, provide the formal reason.

- The matrix is TUM: each column contains at most 2 non-zero elements and these are of opposite signs. [TUM property is preserved under transpose- can you think why?]
- constraints:

$$x_1 - x_2 - x_3 \leq 1$$

$$-x_1 + x_4 \leq 1$$

$$x_1 - x_4 \leq 1$$

$$x_3 \leq 1$$

- Consider $f(x_1, x_2) = \begin{cases} (x_1 - 3)^2 + (x_2 - 2)^2 & x_1, x_2 \geq 0 \\ x_1 + x_2 & \text{otherwise} \end{cases}$
demonstrate how you can use gradient descent to find a local maxima/minima for f . Show up to 2 iterations.
- $\nabla f = \begin{bmatrix} 2(x_1 - 3) \\ 2(x_2 - 2) \end{bmatrix}$ Let $t = -0.5$ and $x^{i=1} = (1, 1)$. Then
 $x^{i=2} = x^{i=1} - 0.5 \begin{bmatrix} 2(x_1 - 3) \\ 2(x_2 - 2) \end{bmatrix}$ which implies
 $x^{i=2} = (-1, 0)$. Check $f(x^{i=2}) < f(x^{i=1})$ for minimization and repeat.

Envelope theorem [Q6 PS 5]

Consider the following maximization problem:

$$\max ax_1 + bx_2 - x_1^2 - x_2^2 + x_1x_2$$

where $a > b > 0$. Find the impact of a on the optimal value using envelope theorem. Show all the steps.

- $\frac{\delta f}{\delta a} = x_1$ at $x_1 = x_1^*$ using envelope theorem. Now $\frac{\delta f}{\delta x_1} = a - 2x_1 + x_2 = 0$ at $x_1 = x_1^*$, which implies $x_1^*(a) = \frac{a + x_2}{2}$. Therefore, $\frac{\delta f}{\delta a} = x_1 = \frac{a + x_2}{2}$ at $x_1 = x_1^*$.

Sensitivity analysis [PS 5, Q7]

- Let $f(x) = ax^2 + c$, $a, c > 0$ and $f_i(x) = x - 3$. Suppose that $f_i(x) \leq u$ is a constraint for minimization of f , for a given $u \in R$. Write down the convex minimization problem. Find out the change in the optimized value of f that would occur if the value of u becomes smaller.
- Convex minimization problem:

$$\begin{array}{ll} \text{minimize} & ax^2 + c \\ \text{s.t.} & x - 3 \leq u \end{array}$$

The optimal value at $u = 0$: $f(x = 0) = c$. Note that using Lagrangian, $\lambda < 0$ so our KKT sufficiency cannot be directly applied. Note that the constraint $f_i(x^* = 0) = -3 < 0$, it is therefore inactive. A small change in u will not affect the optimal value. If constraint was active, $\lambda = \frac{-\delta p^*(0, 0)}{\delta u}$. In this case it is simply $\lambda = 0$.